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PI Name:	Hammond, Timothy M.D.		
Project Title:	Yeast Colony Survival in Microgravity Depends on Ammonia Mediated Metabolic Adaptation and Cell Differentiation -Flight		
Division Name:	Space Biology		
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PI Email:	timothy.hammond@duke.edu	**Fax:**	FY 9193238644
PI Organization Type:	NON-PROFIT	**Phone:**	9194285055
Organization Name:	Institute of Medical Research/Department of Veterans Affairs		
PI Address 1:	Institute For Medical Research, Inc.		
PI Address 2:	508 Fulton St, Building 15, Room 109		
PI Web Page:			
City:	Durham	**State:**	NC
Zip Code:	27705-3875	**Congressional District:**	1
Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
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Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
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Key Personnel Changes/Previous PI:	May 2015 report: Added Drs. Corey Nislow and Guri Giaever as Co-Is.		

COI Name (Institution):	Birdsall, Holly M.D., Ph.D. (Baylor College of Medicine) Nislow, Corey Ph.D. (University of British Columbia) Giaever, Guri Ph.D. (University of British Columbia)
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Task Description:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound-target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. The modulation of yeast colonies, especially the redox potential, allows study of different drug pathways. This application investigates microbial metabolic adaptation and cell differentiation to the multiple stimuli encountered in space flight environments, using the yeast Saccharomyces cerevisiae as a model system. We have shown that microgravity activates a unique set of genes in yeast. Our previous work has largely defined yeast cellular stress defense response to microgravity, but stress defense responses are only involved in the survival of yeast in liquid cultures. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. These two populations have different metabolic pathways and establish a cycle of nutrients that allows the U cells to continue to proliferate at the leading edge. Without ammonia signaling, the entire colony dies. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. We postulate that the flow of nutrient and ammonia signals will differ in microgravity where there is little to no convection. We hypothesize that yeast colony survival in microgravity depends on ammonia mediated metabolic adaptation and cell differentiation rather than on stress defense. To test our hypothesis, we will identify the mechanisms mediating colony stratification into U and L cells under multiple gravitational environments: microgravity emulated by Rotating Wall Vessel and Random Positioning Device, static one g cultures, and ultimately microgravity of space flight. Our experimental model will compare liquid and solid phase yeast cultures and make use of available mutant strains of yeast with deletion of the enzymes involved in ammonia signaling and the connected alternative metabolism activation, enzymes involved in stress responses, and the Msn4 promoter and SSA4 gene that we have previously identified as mediating response to microgravity. Relevant metabolic pathways will be studied using green fluorescent protein-linked reporter genes that can be analyzed by confocal microscopy of intact colonies, and flow cytometry of cells divided into L and U populations by sucrose density centrifugation. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
	The yeast Saccharomyces cerevisiae has broad genomic homology to human cells and is simple to grow and genetically manipulate (Lee et al. 2014; Kachroo et al. 2015; Nislow et al. 2015). This has made it a powerful model organism with biomedical applications to understand mechanisms of cell-to-cell interactions, cell survival, and aging (Cap et al. 2009a; Ayer et al. 2014; Herker et al. 2004). To survive, yeast in liquid cultures must use strategies that are very different from that used by yeast in agar-based colonies (Cap et al. 2009a). Survival in liquid culture is dependent on stress defense whereas survival in a colony is dependent on metabolic adaptation and stratification of the population in response to ammonia-mediated signaling (Cap et al. 2009a). Both modes of culture have applications for biomedical research. Chemical genomic assays using a deletion series of Saccharomyces cerevisiae grown in rich liquid media have provided powerful methods to identify the mechanism of action of known drugs and novel small molecules in vivo including chemotherapy anti-cancer agents (Smith et al. 2010). On the other hand, stratification and divergent metabolic adaption of yeast within a colony on solid agar models many of the changes seen within solid tumors (Cap et al. 2009a, 2010, 2012a, d). Yeast is also a popular model for studies of cellular responses to microgravity and microgravity simulations. Multiple flight experiments have verified that S. cerevisiae remains fully viable, and responds to the microgravity environment with changes in metabolism [e.g., increase in phosphate uptake (Berry and Volz 1979) and phenotype [e.g., increase in number and distribution of bud scars (Walther et al. 1996)]. Studies of yeast in space have typically been conducted with liquid cultures. The responses of liquid culture yeast to real (e.g., space flight) microgravity and simulated microgravity include changes in reactive oxygen species, apoptosis, as well as shear defense mechanisms (Coleman et al. 2007, 2008a, b; Hammond et al. 1999; Johanson et al. 2002; Nislow et al. 2015). We have used a deletion series strategy, similar to chemical genomic assays, to demonstrate that mitochondrial and ribosomal redox gene pathways play a predominant role in the responses of yeast colonies to microgravity (Nislow et al. 2015). This genetic modulation of yeast colonies in real and simulated microgravity is dependent, at least in part, on shear stress promoters, apoptosis, and reactive oxygen species (Coleman et al. 2008a). The first steps in this initiative were to validate assays for reactive oxygen species and glutathione in Saccharomyces cerevisiae during microgravity simulation. The effects of space flight on yeast have high concordance with agents that induce a very low intracellular redox state, and induce a massive efflux of glutathione. These results raise important issues. Can the reduced redox state during space flight be reproduced and modulated in ground-based simulations? Will this allow definition of unique drug pathways, as a low redox state mirrors the electrophilic properties of mitochondria, where many drugs are metabolized? Unfortunately, assays for redox status and its major cellular determinant, glutathione, are diverse, often cell type specific, and we lack an accepted probe set for yeast

studies. This paper validates fluorescent probes for glutathione and reactive oxygen status in yeast to support mechanistic studies of microgravity and drug metabolism. The plethora of fluorescent reagents for reactive oxygen species and glutathione can be bewildering, and makes head-to-head comparisons of all the alternatives impractical. These reagents measure the physiological milieu of reactive oxygen species and diverse thiols, rather than specific individual molecules. We report that in yeast, monochlorobimane (mBCL) and 2',7'-dichlorodihydrofluorescein diacetate (DC-FDA) are suitable for fluorometric and flow cytometry studies of glutathione and reactive oxygen species, respectively. Both dyes have low background fluorescence, predictable loading, good retention, and are non-toxic to Saccharomyces cerevisiae. Both dyes show concordance with other fluorescent and biochemical assays of reactive oxygen species.

The second step in this initiative was to establish the timing, optimal media (rich versus poor) a dynamic range of rotational interventions for low redox potential in giant yeast colonies. Giant yeast colonies develop a low redox potential, which mimics the electrophilic milieu of both the mitochondrial drug metabolizing compartment, and the hypoxic core of many tumors. The major metabolic mediators of this low redox potential include ATP, glutathione, NAD+/NADH, and NADP+/NADPH. Ammonia signaling is the critical mechanism that induces stratification of the giant yeast colonies to allow a low redox potential. Comparison of two powerful investigative models for drug pathways using Saccharomyces cerevisiae have been compounded by the use of different growth media and stimuli to the system. Chemogenetic profiling, which uses a pool of yeast deletion mutants to determine survival changes, is heavily slanted to the use of rich media. Giant yeast colonies studies are heavily slanted to the use of poor media. The current study answers three questions. First, what are the differences in redox potential and its major metabolic mediators in giant yeast colonies over time? Metabolic status is assayed as glutathione content, redox potential, ATP content, and ratios of NAD to NADH and NADP to NADPH. Second, how does rich versus poor media affect physiological responses of giant yeast colonies? Third, how does clinorotation affect redox potential and its metabolic mediators? Clinorotation is of interest because it randomizes the gravitational vector. That in turn should disrupt the convection-medicated ammonia signaling known to drive giant yeast colony stratification and differentiation. This study demonstrates that the low redox potential status in tumors and drug metabolizing areas of mitochondria can be reproduced in giant yeast colonies grown on both poor and rich media. The study defines the time course and effects of middle molecule mediators from glutathione and ATP to the NAD family of mediators. Rotation proved to be a potent way to selectively modulate ammonia signaling, redox potential, and metabolic mediators.

The third step in this initiative was to determine if lack of convection of gaseous ammonia in true microgravity changes the stratification and expansion of in giant yeast models of differentiation and tumorgenesis.

Little is known about the effects of space flight on yeast growing as a colony on agar. On Earth, yeast cells form giant multicellular colonies with characteristic organized morphologies (Cap et al. 2009b, 2012b, c). Around day nine, cells at the base of the colony begin to apoptose and also begin to secrete ammonia. Ammonia signaling induces cells at the top and leading edges of the colony to reprogram their metabolic pathways and divide rapidly, thereby allowing the colony to continue to expand. Yeast colony stratification is indelibly linked to gravity-driven convection, as ammonia production and the gravity driven convection of produced ammonia mediate giant yeast colony differentiation (Cap et al. 2009a; Palkova et al. 1997). It is not known how the absence of convection seen in microgravity might affect this process.

To answer this question, we studied giant yeast colonies formed in true microgravity (i.e., space flight) and in two ground-based simulations of microgravity: rotation and random positioning. Rotating culture devices have been a popular model to mimic some of the physical factors induced by microgravity during space flight (Hammond and Hammond 2001; Lee et al. 2014; Birdsall et al. 2016). Redirecting gravity driven convection, by growing yeast on a vertical rather than horizontal agar slab, disrupts ammonia signaling, altering the redox status of yeast cells and gene expression kinetics during growth of giant yeast colonies (Birdsall et al. 2016). The random positioning machine (RPM), referred to as the 3-D clinostat, is a microgravity simulator based on the principle of ‘gravity-vector-averaging’, which can generate simulated gravity levels from 0g to 1g (Grimm et al. 2014). There is limited data comparing the RPM, simpler 2-D rotation, and true microgravity (Grimm et al. 2014), and controversy as to whether the forces they induce are distinct (Klaus et al. 2004; Murdoch et al. 2013; Wuest et al. 2017).

Metabolic adaption in giant yeast colonies is triggered by the depletion of nutrients in the agar and to enhance this process colonies are typically grown on nutrient-poor agar (Cap et al. 2009a; Palkova et al. 1997). In contrast, studies of yeast in liquid cultures, including chemical genomic assays, tend to use rich media. To evaluate the role of nutrients, we conducted parallel studies of yeast on both nutrient-rich (YPD) and nutrient poor (YE) media.

Three major analyzes were conducted in yeast grown as giant colonies under true and simulated microgravity. First, we measured the expression of a series of genes likely to be affected by microgravity using clones engineered to express enhanced green fluorescent protein (eGFP). Table 1 lists the clones used and the rationale for their selection. ATO3, INO1, SSA4, and RPL34b are known to be upregulated in different strata as yeast differentiate into giant colonies in ground studies (Cap et al. 2012a, d). BEM1 was included as an index of bud scarring, which is known to be altered in liquid cultures of yeast in space flight (Walther et al. 1996). TRR1 was included because giant colony differentiation on the ground and in rotation is accompanied by changes in redox (Birdsall et al. 2016; Cap et al. 2012d) and TRR1 is the key control enzyme in production of glutathione, the most abundant thiol buffering cellular reactive oxygen species (Trotter and Grant 2005). EDC3 was included as it is often constitutively expressed, proving a control for a field change effect (Kshirsagar and Parker 2004). Second, we evaluated the dependence of two stress response genes, SSA4 and RPL34b, on the stress promoters, Msn4 and Sfp1 (Johanson et al. 2002). We have previously shown that in liquid yeast cultures, SSA4 and RPL34b have novel transcription dependence on Msn4 and Sfp1 during space flight that differs from ground-based controls (Coleman et al. 2008a).

Third, we evaluated the viability and induction of apoptosis in the giant yeast colonies and the dependence of these outcomes on the stress promoters Msn4 and Sfp1. The differentiation of giant yeast colonies in terrestrial analyzes is dependent on the promoters Msn4 and Sfp1 (Cap et al. 2012b) but the effects of microgravity on this process are not known.

100 µ l of stationary culture inocula of each yeast strain were placed in 5 ml of fresh YPD or YE media and grown overnight in a shaking incubator at 30°C. Omniwell plates prefilled with 40ml of YE or YPD agar were spotted with 60 µ l of the overnight liquid culture with forty-eight spots per plate. To minimize dehydration after drying the Omniwell lids were sealed to the base with Durapore tape and the plates were placed in plastic Ziploc™ bags. The plates were cooled to 4°C and were divided into five sets of four plates - one set for each of the five growth conditions. Each set contained a full series on YPD agar and a full series on YE agar.

Flight and static ground control plates were sent to Kennedy Space Center in chilled shipping containers. Just prior to flight, working in a cold room, the plates were transferred to barcoded Ziploc™ bag and stacks of eight were placed in Lexan bags and then into plate habitats (PHABs; http://www-bioserve.colorado.edu/; BioServe Space Technologies, CO). The PHAB is a sealed container that allows for gas exchange but meets NASA safety

requirements for levels of containment. (https://). PHABs were placed in cold bags at 4°C, and remained chilled for five days during handoff to the SpaceX-8 rocket. Flight samples were flown to the International Space Station (ISS). Approximately two days post-launch, the growth of the yeast on ISS was initiated by transferring the PHABs to Commercial Generic Bioprocessing Apparatus 4 (CGBA4) (Hoehn et al. 2004) that was set to 30°C (BioServe Space Technologies, CO). After 28 days in flight, CGBA4 was commanded down to +4°C. The following day the PHABs were moved back to cold bags, and returned to Earth the next day on the Dragon capsule. Plates remained chilled for a total of six days during undocking, return to Earth, salvage, and transport to the lab for flow cytometry analysis.

The second set of plates, the static flight controls were loaded into a second set of PHABs at Kennedy Space Center and handled to match the exact timing and temperature profile of the flight samples. Ground controls were conducted with a 24-hour offset relative to the flight samples and the static controls in PHABs followed the same temperature changes as the flown samples.

A third and fourth set of samples were subjected to Wall-Vessel rotation or random positioning (RPM) during growth. The timing of the rotating and RPM cultures matched that of the flight samples with the exception that they were grown at ambient laboratory conditions (25°C) rather than 30°C and they were not placed in PHABs. To control for the difference, an additional set of static ground controls were maintained in plastic bags at 25°C and also not placed in PHABs. Results from the two set of static controls, PHABs/30°C versus no-PHAB/25°C, did not differ (Pearson correlation >95%) so data from the two were pooled in the statistical analyzes. Similarly, we could not detect any differences due to plate position of the replication using any of the modalities studied (Pearson correlation >95%).

We have previously reported on the effects of media and rotation on the viability and redox potential in giant yeast colonies in ground-based studies (Birdsall et al. 2016). In this report, we extend analysis to include giant yeast colonies in the true microgravity of space flight as well as in the microgravity simulation by random positioning (RPM) and 2D rotation (Rotating Wall-Vessel head) positioning. Defining the utility of microgravity and microgravity simulations is confounded by use of different microgravity simulations, different media, and different outcomes. This study is a systematic examination of the effects of true microgravity versus common microgravity simulations, and nutrient rich media versus nutrient poor media, on major mechanistic mediators including shear stress promoters Msn4 and Sfp1, reactive oxygen species, and apoptosis. This lays the ground work for future yeast experiments to study cell-to-cell interactions, cell signaling, cell survival, colony differentiation and aging, mechanisms of drug action, and characteristics of tumor generation.

Cell death by necrosis and apoptosis increased in flight as shown by multiple modalities: uptake of propidium iodide up, expression of caspase, and binding of annexin V. Induction of apoptosis by simulated microgravity has been seen with other cell types including human lymphocytes (Ward et al. 2006). This increased cell death was greatly increased by deletion of either of the stress responsive promoters Msn4 or Sfp1. The finding of both necrotic and apoptotic cell death is further evidence for activity of both ammonia signaling and stress defense pathways, as ammonia signaling utilizes mostly caspase independent pathways (Vachova and Palkova 2005) and stress defense is mediated mostly by apoptosis as well as autophagy and necrosis (Farrugia and Balzan 2012).

The changes in INO1 and SSA4 genes are significant in both random positioning with gravity vector averaging and rotation techniques but oppositely directed to the space flight changes, underscoring the uniqueness of the effects of true microgravity. During culture on the ISS, yeast cultures are also exposed to increased radiation (Nislow et al. 2015). Our previous studies using yeast deletion series in liquid culture to evaluate the effects of space flight revealed two dominant responses: radiation/DNA repair and reactive oxygen pathways (Nislow et al. 2015). We conclude that the ISS provides a unique suite of stimuli with distinct set of responses: the question is how to translate these properties to i) further NASA’s program objectives, ii) support commercial development, and iii) address unmet clinical needs. We found differences in the responses of yeast in the rotating wall vessel versus random positioning models in both ammonia-dependent and stress-dependent changes, providing clarity to the controversy as to whether the random positioning device is a form of rotation, or a distinct stimulus (Klaus 2001; Klaus et al. 2004). These differences included gene expression changes mediated by shear stress promoters, redox status, and apoptosis. Hence, we conclude that the forces induced by random positioning and rotation are distinct. Many resources are available for scientists seeking to study the effects of gravity on biologic systems (Brungs et al. 2016; Frett et al. 2016) and researchers should be mindful that fluidity of the culture milieu and shear stress at the membrane surface are likely to be important variables affecting how cells sense gravity (Kohn et al. 2017; van Loon 2008). In our model system composed of a near-solid colony of yeast, receiving nutrients from an agar gel at its base, and using a gas, ammonia, as one means of driving differentiation within the colony, space flight, rotating wall vessel and random positioning generate different effects. In free-floating liquid systems such as U937 cells growing in suspension (Bradamante et al. 2006), HUVEC cells adherent to Cytodex beads (Bradamante et al. 2006), or thyroid cells growing as spheroids (Warnke et al. 2016) rotating wall vessel, random positioning machines, and space flight had very similar effects.

The ability of physical methods to manipulate yeast colony redox, apoptosis, and stress defense status is important at multiple levels of organismal biology. Gene deletion, although targeted to a single locus, will often cause follow-on effects in multiple integrated pathways, especially when promoters are deleted (Park et al. 2011). Further, there are clinically important applications. Analysis of yeast survival pathways offers powerful techniques to address drug metabolic mechanisms for repurposing of pharmaceuticals (Lee et al. 2014), including identification of unique targets (Blackman et al. 2012) and buffering pathways required to respond to drugs. Further, yeast colony differentiation recapitulates many of the stratification mechanisms of tumorigenesis (Cap et al. 2012a, d; Birdsall et al. 2016). In conclusion, by allowing graded manipulation of a combination of convection and ammonia signaling, while avoiding secondary effects of gene deletion, these physical techniques are primed to substantially contribute to our understanding of mechanisms of drug action, cell aging, and colony differentiation. The fourth step in this initiative was to determine the gene pathways, which mediate space based changes in giant yeast colony differentiation as a tumorogenesis model.

Our data shows that space flight on the ISS appeared to have a unique pattern on DNA damage. The logical next step is to combine this genome sequencing data with functional data in order to investigate the biological consequence(s) of deleting a particular genes and the subsequent accumulation of mutations.

The hypothesis to be tested is that yeast double deletion pools can be used to identify genes conferring resistance to space/cosmic radiation.

The traditional approach is to create double mutants that contain combinations of each mutation of interest (the query mutation) with a mutation in a second gene (Dixon et al., 2009). This experimental approach exploits the phenomenon of synthetic lethality in which a combination of mutations in two or more genes leads to cell death, whereas a mutation in only one of these genes is not, in itself, lethal. Synthetic lethality is currently of great interest in the study of cancer. Synthetic lethality can help us understand how cells become cancerous by overcoming the redundant cellular quality control mechanisms. It can also be a potential way to selectively kill cancer cells by identifying drugs that can target and inactivate synthetic lethal genes that act in concert with

Task Progress:

cancer-driver genes. Our previous radiation work in space has not addressed synthetic lethality at all.

We generated double mutant libraries in which the mutated DNA repair gene (mutation #1) is paired with a homozygous deletion of another yeast gene (mutation #2). Each pool contains approximately ˜ 4600 double-mutant strains, representing deletion of each of the nonessential genes in the yeast genome combined with mutation #1. In “fitness profiling” the mutants were grown and the relative quantity of each clone was measured (Roemer et al., 2012; Smith et al., 2010). The samples were flown cold to the ISS, allowed to grow while exposed to cosmic radiation on station, and returned to ground. Recent innovations in sample preparation allowed us to fly these experiments at ambient temperatures, which required minimal crew input to initiate the experiment, and produced a small payload. Furthermore, because the experiment was limited by nutrient availability (e.g., the yeast grow until the nutrients are depleted then remain in a quiescent state), the experimental set up is a closed system that did not require any additional intervention or specialized post-experiment storage requirements. Finally, the generation of these double mutant pools provides sufficient stock material for ~5000 ground based follow-on experiments using the sample aliquots, and can be stored indefinitely.

To prepare the second mutation in each clone, both copies of each of the ˜ 4600 nonessential genes in the yeast genome is removed by genetic engineering. The mutated clone is able to thrive so long as there is no external stressor targeting that particular gene’s protein. However, in the presence of the stressor (e.g., cosmic radiation), the clone is overwhelmed and grows less well compared to clones with irrelevant mutations. To discriminate the clones in the series, each clone’s deleted gene has been replaced with a unique DNA bar code. The entire series can be grown together in the presence of the stressor, the DNA extracted, and the numbers of each clone quantified by high throughput sequencing or hybridization to gene chips (Smith et al., 2010). The ‘clone-identifying’ barcode is unique to each clone, but the insert is flanked by up and down DNA sequences that are identical in every clone so that the extracted DNA for all clones can be amplified together. All of the techniques proposed have been used and validated in space flight (Johanson et al., 2007; Nislow et al., 2015).

Multiple double mutation pools were prepared corresponding to sentinel DNA-sensitive mutant strains. These pools were made robotically, rapidly, and inexpensively (Giaever and Nislow, 2014). The result of the chemogenomic assay is a quantitative metric, termed "fitness", presented as a list of genes ranked in order of their importance for growth and survival in the presence versus the absence of the stressor (space flight/cosmic radiation vs ground). Accordingly a ranked list of all genes in the genome was generated for each experiment and then compared using gene set enrichment analysis (GSEA) to define those pathways and processes that were affected by growth in microgravity using Cytoscape (Wallace et al., 2011).

Detecting differentially fit genes from next generation sequencing or microarray data involves a set of statistical tradeoffs. Standard methods look for genes by relying on elementary test statistics using data from one gene at a time (such as t-tests, or contrasts from ANOVA models). Gene lists are obtained by filtering statistics by attempting to control overall type-I error, or false discovery rate, or by filtering using a user specified cutoff level. The tradeoff between controlling Type I errors, due to the large number of comparisons, and maintaining power can be handled in a variety of ways. These methods include but are not limited to False Discovery Rate analysis including Storey statistics, Bayesian analysis, and many other mapping techniques.

Specific comparisons provide internal controls. We were able to identify most of the nonessential genes previously shown to be involved in nucleotide excision repair, in cell cycle checkpoints, in homologous recombination, and in post replication repair after other forms of radiation damage. This validates the space-based assay by acting as a positive control.

Second, comparison is being made of the ISS data to existing data sets in this model on UV irradiation, alkayating radiomimetic agents, and our prior ISS data. Especially in the deletion pool using the wild type background, this should elucidate if the changes in space have unique pathways.

The experiments performed in space flight and ground-based controls directly address Human Research Program and Space Biology Science Plan's research goals, objectives, and guiding questions that underlie our project's unifying hypothesis: "Microgravity-induced changes in yeast are induced, at least in part, through changes in density driven ammonia convection and the subsequent effects on Msn4- and Sfp1-dependent genetic and cellular differentiation of giant yeast colonies." Each experiment is identifying specific target genes and the transcription factors that mediate the changes using reporter synthetic genetic array methodology or gene expression.

The Micro 9 experiment flew on SpaceX-8, spending one month on the ISS. There were zero defects in either the ISS or parallel ground control experiments. All four experiments included static and rotated ground controls. Experiments 3 and 4, ground controls with a random positioning machine, were also conducted.

Experiment 1. Evaluate the role of ammonia and reactive oxygen species in the response to space flight using synthetic gene array technology. The yeast deletion series were prepared in a background of suppressed thioredoxin reductase activity (TRR1DAmp). Thioredoxin reductase is the key control enzyme in production of glutathione, the most abundant thiol buffering cellular reactive oxygen species. Yeast deletion series were also prepared in a background of deleted SOK2, so that they are unable to synthesize ammonia, a key signaling molecule in the biochemical differentiation of giant yeast colonies. Responses in these series were compared to yeast deletions series in a wild type background.

Individual clones were spotted on rich agar for fitness profiling. Pools of each series were also spotted on nutrient-rich and nutrient-poor agar as nutrient status affects generation of reactive oxygen species. The agars were prepared with and without addition of the DNA-damaging agent 0.1 mM doxorubicin, as an additional stressor.

Experiment 2: Evaluate the response to space flight with radiation-sensitive clones. Each of the 26 rad deletion addresses a different DNA damage response with different specific target gene and transcription factor responses in ground-based studies.

Experiment 3: Evaluate the role of Sfp1 and Msn4 transcription factors in the expression of SSA4 and RPL34B by giant yeast colonies grown in real and emulated microgravity. eGFP-expressing SSa4 (YER103W) and RPL34B clones with and without deletion of Msn4 & Sfp1 transcription factors were used. RPL34B and SSa4 were selected because we had previously found that their dependence on Msn4 and Sfp1 is altered in microgravity in liquid cultures. Specific clones were: Ssa4 (YER103W); Sfp1 delta Ssa4; Msn4 delta Ssa4; RPL34B (YIL052c); Sfp1 delta RPL34B; Msn4 delta RPL34B. Yeast were evaluated for shear stress gene expression, cell death, apoptosis, and reactive oxygen species by flow cytometry.

Experiment 4: Evaluate the expression of genes in giant yeast colonies grown in real and simulated microgravity. eGFP-expressing yeast clones were spotted on nutrient rich and nutrient poor agar. Yeast were evaluated for shear stress gene expression, cell death, apoptosis, and reactive oxygen species by flow cytometry.

The rationale for the clones selected is as follows: ATO3, INO1, SSA4, and RPL34b are known to be upregulated in different strata as yeast differentiate into giant colonies in ground studies. BEM1 was included as an index of bud scarring, which is known

to be altered in liquid cultures of yeast in space flight. TRR1 was included because giant colony differentiation on the ground and in rotation is accompanied by changes in redox and TRR1 is the key control enzyme in production of glutathione, the most abundant thiol buffering cellular reactive oxygen species. EDC3 was included as it is often constitutively expressed, proving a control for a field change effect.

For all samples, DNA has been extracted, the reporting bar codes have been amplified by PCR, and sequencing libraries prepared. Next generation sequencing has been completed, as has initial analyses by Gene Set enrichment Analysis (GSEA). Visualization and probability modeling by Cytoscape and ClueGO is ongoing.

All OMICS data will be annotated and deposited in the NASA GeneLab portal.

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